

<p align="center"> </p> <h1 align="center"> 🌊 TechTide Swarm 357</h1> <p align="center"> 357 Specialized Claude AI Agents That Run Like a Full Digital Company </p> <p align="center"> Built on Anthropic's Claude Agent SDK · Opik/Comet Observability · 6-Layer Business Architecture </p> <p align="center"> </p> <p align="center"> Architecture • Agent Catalog • Quickstart • Observability • Management Layer • Docs </p>

Most people use Claude like a simple tool. The real power is building an AI company.

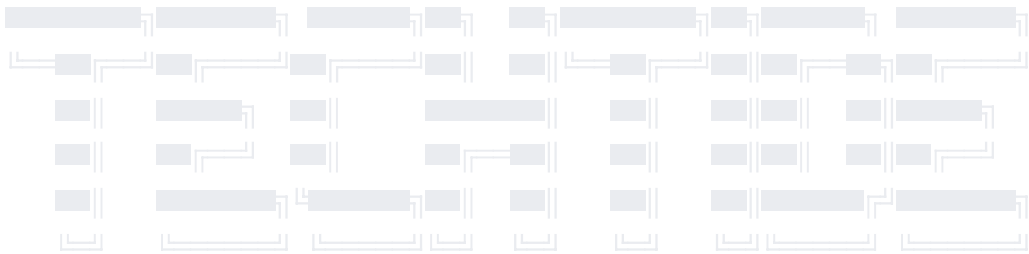
TechTide Swarm 357 is a production-grade, open-source agent ecosystem that deploys **357 specialized Claude AI agents** across 6 business layers — with full observability, evaluation pipelines, and a 10-agent management layer that orchestrates the entire swarm.

Every agent has **one job**. Every trace is **logged**. Every decision is **evaluated**. Every dollar spent is **tracked**.

This isn't a framework. It's a fully operational AI workforce.

⚡ See It in 30 Seconds

```
bash
pip install techtide-swarm-357
swarm init
swarm demo
```



⚡ S W A R M 3 5 7 ⚡

🖥 DEMO MODE – Watch 5 agents work a real task

🌐 Swarm Pipeline

```
├─ 🌐 Stage 1: Research (parallel)
│   ├── research-market-trends-001 → Market analysis
│   └── seo-keyword-research-001 → Keyword discovery
├─ 🛠 Stage 2: Create (parallel)
│   ├── mktg-content-blog-001 → Blog post
│   └── sales-outreach-email-001 → Outreach email
└─ 📄 Stage 3: Support
    └── support-kb-writer-001 → FAQ article
```

✅ 5 agents · 5 layers · 18.4s · \$0.14 total · avg eval: 0.90

vs How We Compare

Feature	Swarm 357	OpenClaw	SuperClaude	Ruflo	claude-skills
Purpose	Full AI workforce	Personal assistant	Dev framework	Orchestration	Skill library
Pre-built agents	357	0 (SOUL templates)	16	~50	220 skills
Business layers	6 layers	—	—	—	—
Management meta-agents	10	—	—	—	—
Observability	Opik native	Plugin	—	Custom	—

Feature	Swarm 357	OpenClaw	SuperClaude	Ruflo	claude-skills
Cost control	Per-agent budgets	—	—	Model routing	—
Eval pipeline	LLM-as-Judge + CI gate	—	—	—	—
Self-healing memory	3-layer + autoDream	Basic memory	—	Vector memory	—
Security gates	23-point bash validation	User-managed	—	—	—
ULTRAPLAN (deep planning)	✅ Opus offload	—	—	—	—
COORDINATOR_MODE	Research→Synth→Impl	—	—	Swarm routing	—
Feature flags	17 flags	—	—	—	—
Claude Agent SDK native	✅	✅	❌ (config only)	✅	❌
License	MIT	MIT	MIT	MIT	MIT

TL;DR: Other repos give you tools. We give you the entire company.

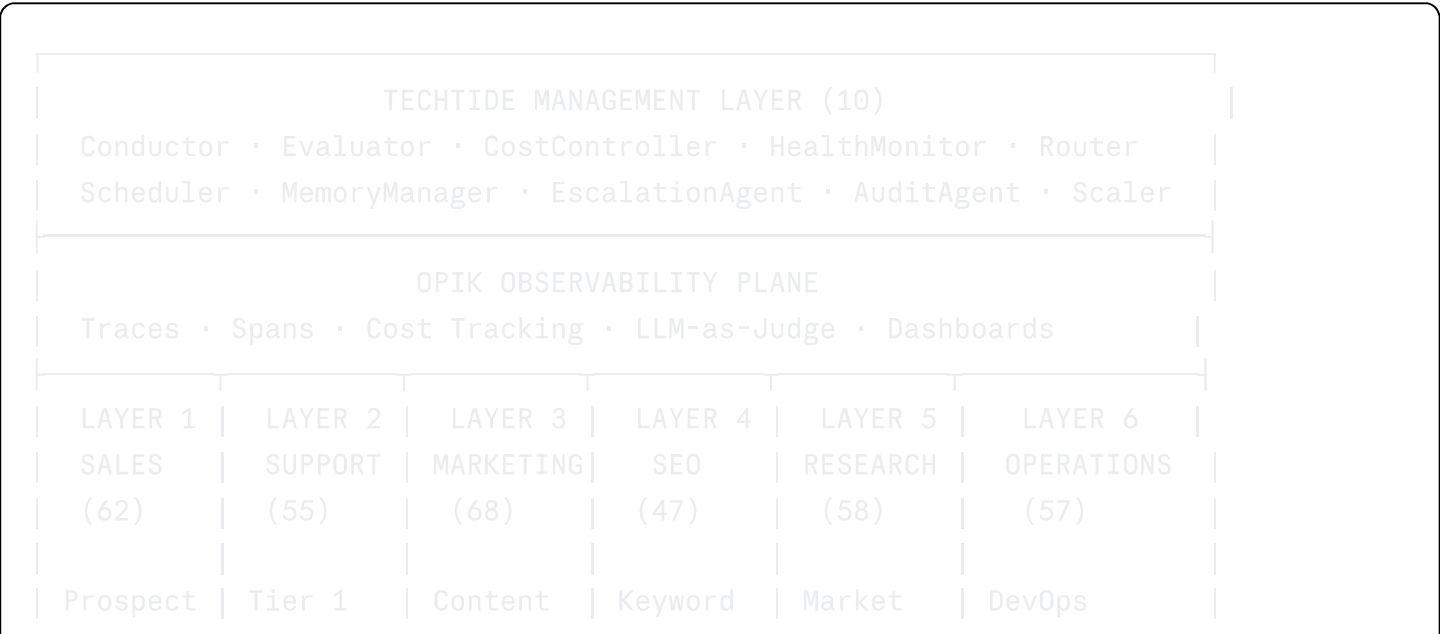
 Reverse-Engineered from Claude Code Internals

When Claude Code's source leaked on March 31, 2026 (512K lines via npm sourcemap), we studied every architectural pattern and built them into this repo:

Pattern from the Leak	Our Implementation
Three-layer self-healing memory (MEMORY.md → Topic Files → Transcripts)	<code>memory/store.py</code> — full 3-layer system
autoDream background consolidation (forked subagent, 4 phases)	<code>DreamEngine</code> with contradiction detection
Memory-as-Hint protocol ("verify before trusting")	Baked into every agent's system prompt
COORDINATOR_MODE (Research → Synthesis → Implementation)	<code>CoordinatorMode</code> with typed phase pipeline
ULTRAPLAN (remote Opus planning, 30-min sessions)	<code>UltraPlan</code> class with structured output
23-point bash security validation	<code>BashSecurityGate</code> — all 23 checks
44+ feature flags	<code>FeatureFlagRegistry</code> — 17 production flags
Five context compaction strategies	<code>ContextCompactor</code> — all 5 strategies
Append-only audit trails (agent cannot self-erase)	<code>TranscriptArchive</code> + <code>AuditAgent</code>

Full technical breakdown: [docs/REVERSE_ENGINEERING.md](#)

Architecture





The 6 Business Layers

Layer	Agents	Description
 Sales	62	Full pipeline: prospecting → qualification → outreach → demo → close → retention
 Support	55	Tiered support: auto-response → escalation → knowledge base → feedback loops
 Marketing	68	Content engine: copywriting → social → email → ads → brand → analytics
 SEO	47	Full stack: keyword research → audits → link building → technical → local SEO
 Research	58	Intelligence: market analysis → competitor tracking → trend detection → patents
 Operations	57	Backend: DevOps → finance → HR → legal → compliance → project management

Total: 347 specialized agents + 10 management agents = 357

 The Management Layer (TechTide 10)

The brain of the swarm. 10 meta-agents that don't do business work — they manage the agents that do.

```
python
```

```
from techtide_swarm import ManagementLayer

mgmt = ManagementLayer()

# The Conductor routes tasks to the right agents
await mgmt.conductor.dispatch(
    task="Write a LinkedIn post about our Q1 results and optimize it for SEO",
    # Conductor auto-selects: ContentWriter → SEOOptimizer → LinkedInPublisher
)

# The Evaluator scores every agent output
await mgmt.evaluator.score(
    agent_id="content-writer-003",
    output=draft,
    metrics=["relevance", "brand_voice", "engagement_potential"]
)

# The CostController kills expensive runs
mgmt.cost_controller.set_budget(
    layer="marketing",
    daily_limit_usd=50.00,
    model_preference="sonnet" # Use Opus only for complex tasks
)
```

Management Agent	Role
Conductor	Task routing, agent selection, workflow orchestration
Evaluator	Output scoring via Opik LLM-as-Judge metrics
CostController	Budget enforcement, model selection, token optimization
HealthMonitor	Agent uptime, error rates, latency tracking
Router	Load balancing, failover, queue management
Scheduler	Cron jobs, heartbeats, recurring workflows
MemoryManager	Cross-agent context, knowledge graph maintenance
EscalationAgent	Human-in-the-loop escalation, approval flows
AuditAgent	Compliance logging, decision audit trails
Scaler	Auto-scaling agent instances based on demand

 Quickstart

Prerequisites

- Python 3.10+
- Claude Agent SDK (`pip install claude-agent-sdk`)
- Opik (`pip install opik`)
- An Anthropic API key

Install

```
bash
```

```
git clone https://github.com/TechTideAI/swarm-357.git
cd swarm-357
pip install -e ".[dev]"

# Configure
cp .env.example .env
# Add your ANTHROPIC_API_KEY and OPIK_API_KEY
```

Deploy Your First Agent

```
python

import anyio
from techtide_swarm import Agent, AgentConfig, SwarmRegistry

# Create a sales prospecting agent
config = AgentConfig(
    name="prospect-researcher-001",
    layer="sales",
    role="prospect_researcher",
    soul="templates/soul/sales/prospect_researcher.md",
    tools=["WebSearch", "Read", "Write"],
    model="sonnet",
    max_turns=15,
    budget_limit_usd=0.50,
)

agent = Agent(config)

async def main():
    result = await agent.run(
        "Research the top 10 AI automation agencies in Ohio "
        "and create a prospect list with contact info, tech stack, "
        "and potential pain points."
    )
    print(result.output)
    print(f"Cost: ${result.cost_usd:.4f}")
    print(f"Opik Trace: {result.trace_url}")

anyio.run(main)
```

Deploy the Full Swarm

```
python

from techtide_swarm import Swarm

swarm = Swarm.from_config("config/swarm.yaml")
await swarm.boot() # Initializes all 357 agents

# Run a cross-layer workflow
result = await swarm.execute(
    "Launch a complete go-to-market campaign for our new AI training product. "
    "Research the market, create content, set up SEO, build email sequences, "
    "and prepare sales collateral."
)

# This triggers agents across ALL 6 layers:
# Research → Market analysis, competitor audit
# Marketing → Content creation, email sequences, ad copy
# SEO → Keyword research, on-page optimization
# Sales → Collateral, pitch decks, outreach sequences
# Operations → Project tracking, budget allocation
```

Observability

Every agent call flows through Opik. Zero exceptions.

```
python
```

```
from techtide_swarm.observability import SwarmTracer

tracer = SwarmTracer(
    project="techtide-swarm-357",
    # Opik Cloud or self-hosted
    opik_url="https://www.comet.com/opik",
)

# Every agent automatically traces:
# ✓ Input/output pairs
# ✓ Token counts & cost per request
# ✓ Latency (time-to-first-token, total)
# ✓ Tool calls and results
# ✓ Subagent delegation chains
# ✓ LLM-as-Judge evaluation scores
# ✓ Error traces with full context
```

Built-in Evaluation Metrics

Metric	Type	Description
answer_relevance	LLM-as-Judge	Does the output match the task?
hallucination	LLM-as-Judge	Did the agent fabricate information?
brand_voice	Custom	Does output match company voice guidelines?
task_completion	Heuristic	Did the agent complete all subtasks?
cost_efficiency	Computed	Output quality per dollar spent
latency_budget	Threshold	Did agent finish within time budget?
tool_accuracy	Heuristic	Were tool calls necessary and correct?
escalation_quality	LLM-as-Judge	Was human escalation appropriate?

Dashboard

bash

```
# Launch the Opik dashboard locally
./scripts/launch-dashboard.sh

# Or connect to Opik Cloud
opik configure --api-key YOUR_KEY --workspace techtide
```

Agent Catalog

Every agent is defined by a **SOUL.md** file (personality, constraints, expertise) and a **SKILL.md** file (tools, workflows, capabilities).

Sales Layer (62 agents)

<details> <summary>Click to expand full Sales agent list</summary>

#	Agent ID	Role	Description
1	sales-prospect-researcher-001	Prospect Research	ICP identification, firmographic data, buying signals
2	sales-prospect-researcher-002	Deep Research	Executive profiles, org charts, decision makers
3	sales-lead-scorer-001	Lead Scoring	Multi-signal scoring: intent, fit, timing, budget
4	sales-lead-scorer-002	Behavioral Scoring	Website visits, content engagement, email opens
5	sales-outreach-email-001	Cold Email	Personalized cold outreach with dynamic variables
6	sales-outreach-email-002	Follow-up Email	Sequence management, timing optimization
7	sales-outreach-email-003	Breakup Email	Final touch sequences, re-engagement
8	sales-outreach-linkedin-001	LinkedIn DM	Connection requests, InMail, comment engagement
9	sales-outreach-linkedin-002	LinkedIn Content	Thought leadership for social selling
10	sales-outreach-phone-001	Call Script	Discovery call scripts, objection handling
11	sales-qualifier-001	BANT Qualifier	Budget, Authority, Need, Timeline assessment
12	sales-qualifier-002	MEDDIC Qualifier	Metrics, Economic Buyer, Decision Criteria/Process
13	sales-qualifier-003	Champion Builder	Internal advocate identification and enablement
14	sales-demo-prep-001	Demo Preparation	Custom demo scripts, feature mapping to pain points
15	sales-demo-runner-001	Demo Execution	Real-time demo support, Q&A handling

#	Agent ID	Role	Description
16	sales-proposal-writer-001	Proposal Writer	SOWs, pricing models, custom proposals
17	sales-proposal-writer-002	Technical Proposal	Architecture docs, integration plans
18	sales-contract-drafter-001	Contract Drafting	MSAs, NDAs, order forms
19	sales-negotiation-001	Negotiation Coach	Pricing strategy, concession planning
20	sales-objection-handler-001	Objection Handling	Response frameworks for common objections
21-30	sales-retention-*	Retention	Churn prediction, renewal, upsell, cross-sell
31-40	sales-analytics-*	Analytics	Pipeline, forecast, conversion, attribution
41-50	sales-enablement-*	Enablement	Training, battlecards, competitive intel
51-62	sales-territory-*	Territory	Account planning, territory mapping, routing

</details>

Support Layer (55 agents)

<details> <summary>Click to expand full Support agent list</summary>

#	Agent ID	Role	Description
1-10	support-tier1-*	Tier 1 Auto	FAQ, password reset, status checks, routing
11-20	support-tier2-*	Tier 2 Technical	Debugging, configuration, integration issues
21-25	support-tier3-*	Tier 3 Escalation	Complex issues, engineering handoff
26-35	support-kb-*	Knowledge Base	Article generation, search, updates
36-40	support-feedback-*	Feedback	NPS, CSAT, sentiment analysis
41-45	support-onboarding-*)	Onboarding	New customer setup, training
46-50	support-proactive-*	Proactive	Health checks, usage alerts, tips
51-55	support-analytics-*	Analytics	Ticket trends, resolution time, agent performance

</details>

Marketing Layer (68 agents)

<details> <summary>Click to expand full Marketing agent list</summary>

#	Agent ID	Role	Description
1-12	mktg-content-*	Content	Blog posts, whitepapers, case studies, ebooks
13-20	mktg-social-*	Social Media	LinkedIn, Twitter/X, scheduling, engagement
21-28	mktg-email-*	Email	Newsletters, drips, triggered sequences
29-36	mktg-ads-*	Advertising	Google, Meta, LinkedIn ad copy & optimization
37-44	mktg-brand-*	Brand	Voice guidelines, messaging, positioning
45-52	mktg-video-*	Video	Scripts, storyboards, captions, thumbnails
53-58	mktg-events-*	Events	Webinars, conferences, promotion
59-64	mktg-analytics-*	Analytics	Attribution, ROI, A/B testing
65-68	mktg-strategy-*	Strategy	GTM planning, campaign architecture

</details>

SEO Layer (47 agents)

<details> <summary>Click to expand full SEO agent list</summary>

#	Agent ID	Role	Description
1-8	seo-keyword-*	Keywords	Research, clustering, intent mapping, gap analysis
9-16	seo-onpage-*	On-Page	Title tags, meta, headers, internal linking
17-22	seo-technical-*	Technical	Crawl audits, schema, Core Web Vitals, sitemap
23-28	seo-content-*	Content SEO	Briefs, optimization, topic clusters
29-34	seo-link-*	Link Building	Prospect lists, outreach, broken link, HARO
35-40	seo-local-*	Local SEO	GMB, citations, reviews, local content
41-44	seo-competitor-*	Competitor	Backlink analysis, content gaps, SERP tracking
45-47	seo-analytics-*	Analytics	Rank tracking, traffic analysis, reporting

</details>

Research Layer (58 agents)

<details> <summary>Click to expand full Research agent list</summary>

#	Agent ID	Role	Description
1-10	research-market-*	Market Intel	TAM/SAM/SOM, market sizing, trends
11-20	research-competitor-*	Competitive	Feature comparison, pricing, positioning
21-28	research-trend-*	Trend Analysis	Emerging tech, industry shifts, signals
29-36	research-customer-*	Customer Research	Personas, journey mapping, interviews
37-42	research-patent-*	Patent/IP	Patent landscape, freedom-to-operate
43-48	research-academic-*	Academic	Paper review, methodology extraction
49-54	research-data-*	Data Analysis	Statistical analysis, visualization
55-58	research-synthesis-*	Synthesis	Cross-source reports, executive briefs

</details>

Operations Layer (57 agents)

<details> <summary>Click to expand full Operations agent list</summary>

#	Agent ID	Role	Description
1-10	ops-devops-*	DevOps	CI/CD, infrastructure, monitoring, deployment
11-18	ops-finance-*	Finance	Invoicing, expense tracking, forecasting
19-26	ops-hr-*	HR	Recruiting, onboarding, policy, reviews
27-32	ops-legal-*	Legal	Contract review, compliance, IP
33-38	ops-compliance-*	Compliance	SOC2, GDPR, HIPAA, audit prep
39-46	ops-pm-*	Project Mgmt	Sprint planning, standups, retrospectives
47-52	ops-qa-*	QA	Testing strategies, bug triage, automation
53-57	ops-security-*	Security	Vulnerability scanning, access reviews

</details>

Creating Custom Agents

SOUL.md Template

Every agent's personality and constraints are defined in a SOUL.md:

markdown

Agent: Content Writer 003

Identity

You are a senior B2B content writer specializing in AI/ML technology.
You write for technical decision-makers (CTOs, VPs of Engineering).

Voice

- Direct, no fluff
- Data-driven claims with sources
- Technical depth without jargon overload
- Conversational authority (like explaining to a smart friend)

Constraints

- Never make unsourced claims about market size or statistics
- Always include a clear CTA
- Maximum 1,500 words unless explicitly requested longer
- Follow AP style guide

Tools Available

- WebSearch: For current data and sources
- Read/Write: For file operations
- Bash: For data processing scripts

Evaluation Criteria

- Relevance to brief: weight 0.3
- Technical accuracy: weight 0.3
- Engagement (hook + flow): weight 0.2
- Brand voice alignment: weight 0.2

Register & Deploy

python

```

from techtide_swarm import Agent, AgentConfig

config = AgentConfig(
    name="mktg-content-writer-003",
    layer="marketing",
    role="content_writer",
    soul="templates/soul/marketing/content_writer.md",
    tools=["WebSearch", "Read", "Write"],
    model="sonnet",
    max_turns=20,
    budget_limit_usd=1.00,
    eval_metrics=["relevance", "brand_voice", "hallucination"],
)

agent = Agent(config)
SwarmRegistry.register(agent)

```

Repository Structure

```

swarm-357/
├─ README.md
├─ pyproject.toml
├─ .env.example
├─ config/
│   ├── swarm.yaml           # Full swarm configuration
│   ├── layers/              # Per-layer configs
│   ├── models.yaml          # Model routing rules
│   └─ budgets.yaml          # Cost limits per layer/agent
├─ src/
│   ├── techtide_swarm/
│   │   ├── __init__.py
│   │   └─ core/
│   │       ├── agent.py     # Base Agent class (wraps Claude Agent SDK)
│   │       ├── config.py    # AgentConfig, SwarmConfig
│   │       ├── registry.py  # SwarmRegistry - agent discovery & lifecycle
│   │       ├── router.py    # Task → Agent routing engine
│   │       └─ types.py      # Shared types and protocols
│   └─ orchestration/
│       ├── conductor.py     # Multi-agent workflow orchestration
│       ├── pipeline.py      # Sequential/parallel agent pipelines
│       └─ swarm.py          # Swarm boot, execute, shutdown

```

```
| | | └─ management/
| | | |   └─ conductor.py    # TechTide Conductor agent
| | | |   └─ evaluator.py    # Output evaluation agent
| | | |   └─ cost.py         # Budget enforcement
| | | |   └─ health.py       # Health monitoring
| | | |   └─ router.py       # Load balancing
| | | |   └─ scheduler.py    # Cron & heartbeat
| | | |   └─ memory.py       # Cross-agent memory
| | | |   └─ escalation.py   # Human-in-the-loop
| | | |   └─ audit.py        # Compliance & audit
| | | |   └─ scaler.py       # Auto-scaling
| | | └─ observability/
| | | |   └─ tracer.py       # Opik integration
| | | |   └─ metrics.py      # Custom eval metrics
| | | |   └─ dashboard.py    # Dashboard config
| | | |   └─ alerts.py       # Alert rules
| | | └─ evals/
| | | |   └─ judge.py        # LLM-as-Judge evaluator
| | | |   └─ heuristics.py   # Rule-based evaluations
| | | |   └─ datasets.py     # Eval dataset management
| | | |   └─ experiments.py  # A/B experiment framework
| | | └─ memory/
| | | |   └─ store.py        # Persistent memory store
| | | |   └─ vector.py       # Vector similarity search
| | | |   └─ context.py      # Cross-agent context sharing
| | | └─ tools/
| | | |   └─ web.py          # Web search & scraping
| | | |   └─ email.py        # Email send/receive
| | | |   └─ crm.py          # CRM integrations
| | | |   └─ social.py       # Social media APIs
| | | |   └─ analytics.py    # Data & analytics tools
| | | └─ agents/
| | | |   └─ sales/          # 62 agent definitions
| | | |   └─ support/        # 55 agent definitions
| | | |   └─ marketing/     # 68 agent definitions
| | | |   └─ seo/           # 47 agent definitions
| | | |   └─ research/      # 58 agent definitions
| | | |   └─ operations/    # 57 agent definitions
└─ templates/
  └─ soul/                 # SOUL.md templates per role
  └─ skill/                # SKILL.md templates per role
  └─ tool/                 # Custom tool definitions
└─ tests/
  └─ unit/
```

```

├── integration/
├── evals/                # Evaluation test suites
├── scripts/
│   ├── launch-dashboard.sh
│   ├── boot-swarm.sh
│   ├── run-evals.sh
│   └── generate-agents.py    # Scaffold new agents from templates
├── docs/
│   ├── ARCHITECTURE.md
│   ├── AGENT_CATALOG.md
│   ├── CONTRIBUTING.md
│   ├── DEPLOYMENT.md
│   └── EVALUATION.md
├── .github/
│   ├── workflows/
│   │   ├── ci.yml
│   │   ├── eval-gate.yml    # Block merges if eval scores drop
│   └── deploy.yml

```

🔑 Key Differentiators

Feature	TechTide Swarm 357	Others
Business-First Architecture	6 layers mapped to real business functions	Dev-tool focused
357 Production Agents	Pre-built, tested, ready to deploy	Build from scratch
10-Agent Management Layer	Self-managing swarm with conductor, scaler, auditor	Manual orchestration
Opik Native Observability	Every trace, every cost, every eval — automatic	BYO logging
Claude Agent SDK Native	Built directly on Anthropic's official SDK	Wrapper frameworks
Cost Control Built-In	Per-agent budgets, model routing, kill switches	Hope for the best
Eval-Gated CI/CD	Merge blocked if agent quality drops	Ship and pray
SOUL.md System	Declarative agent personality & constraints	Hardcoded prompts

🧡 Contributing

We welcome contributions! See [CONTRIBUTING.md](#) for guidelines.

Priority areas:

- New agent SOUL.md templates
 - Custom evaluation metrics
 - MCP server integrations
 - Dashboard visualizations
 - Documentation & tutorials
-

📄 License

MIT License — see [LICENSE](#) for details.

👥 Team

Built by [Alex Cinovoj](#) and [Shane Spencer](#) at [TechTide AI](#) — Ideas Realized.

🎙️ Catch us on the [Automation Vibes](#) podcast.

<p align="center"> If this repo helps you build your AI workforce, drop a 🌟

 The swarm remembers who believed early. </p>